

BURLINGTON ALAMANCE, NC, USA

WMO: 723174

Lat: 36.047N

Lon: 79.477W

Elev: 188

StdP: 99.09

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 93783

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.5	-18.0	0.8	-3.5	-15.6	1.0	-1.1	9.3	8.6	8.4	7.8	1.9	300	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.7	23.4	33.4	23.3	32.3	23.0	25.3	31.6	24.7	30.6	24.1	29.6	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.8	27.6	23.0	18.1	27.1	22.5	17.6	26.6	78.0	31.6	75.6	30.7	73.3	29.8	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	7.1	5.8	-12.0	36.7	2.7	1.6	-13.9	37.8	-15.5	38.8	-17.1	39.7	-19.1	40.8
			-12.8	26.0	2.5	0.7	-14.6	26.5	-16.0	26.9	-17.4	27.3	-19.2	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	4.5	6.2	10.4	15.8	20.1	24.4	26.1	25.4	22.0	15.8	10.0	6.5
	DBStd	8.74	5.72	5.27	5.32	4.53	3.93	2.72	2.20	2.45	3.24	4.49	4.63	4.95
	HDD10.0	583	189	131	61	7	0	0	0	0	0	7	56	132
	HDD18.3	1880	430	342	249	101	29	1	0	0	8	103	252	366
	CDD10.0	2649	17	24	75	181	314	431	500	477	361	187	56	26
	CDD18.3	906	0	1	5	24	86	182	242	218	119	26	2	1
	CDH23.3	8358	2	7	54	286	772	1723	2347	1986	933	229	18	2
	CDH26.7	3388	0	1	5	65	246	733	1079	863	341	55	1	0
Wind	WSAvg	2.3	2.6	2.8	2.9	2.9	2.4	2.0	1.8	1.7	1.9	2.1	2.2	2.3
Precipitation	PrecAvg	1134	87	69	110	92	85	106	115	104	113	83	79	86
	PrecMax	1646	190	152	221	180	210	309	185	221	346	234	230	168
	PrecMin	816	24	31	28	22	24	36	34	27	10	1	6	28
	PrecStd	180	40	28	47	42	49	64	42	49	84	65	53	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	24.2	27.3	30.7	33.0	35.5	36.5	36.5	34.2	31.0	26.1	22.8
		MCWB	15.8	17.3	16.3	18.5	21.9	23.5	24.2	23.0	21.4	22.4	17.9	17.1
	2%	DB	19.0	21.2	24.8	28.5	31.0	33.8	34.7	34.1	32.3	28.0	22.8	20.0
		MCWB	14.4	14.8	15.5	18.2	21.1	23.0	24.0	23.6	21.8	19.9	15.7	16.2
	5%	DB	17.0	18.5	22.5	27.0	29.4	32.4	33.3	32.7	30.6	26.0	20.9	17.7
		MCWB	13.0	13.2	14.2	17.1	20.6	22.8	23.9	23.3	21.6	18.7	15.8	14.3
	10%	DB	14.1	15.7	19.9	24.6	27.7	30.9	32.1	31.3	28.6	23.8	18.7	15.3
		MCWB	10.0	10.0	13.0	16.3	20.0	22.2	23.5	23.1	21.0	18.4	14.2	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	18.6	18.5	21.3	23.4	25.2	26.1	26.0	24.8	24.1	20.0	19.0
		MCDB	19.7	22.3	23.9	26.2	28.8	32.3	33.0	32.0	29.5	26.0	22.2	20.8
	2%	WB	15.5	16.5	17.1	19.8	22.5	24.4	25.4	25.2	23.8	22.3	18.3	16.9
		MCDB	18.3	19.9	22.3	25.2	28.5	31.0	32.1	31.0	28.5	25.8	21.0	19.1
	5%	WB	13.6	13.9	15.8	18.7	21.7	23.8	24.9	24.5	23.0	20.7	16.9	15.1
		MCDB	16.5	17.5	20.5	24.2	27.6	30.0	31.2	29.9	27.2	24.0	19.7	17.3
	10%	WB	10.7	10.8	14.5	17.6	20.8	23.2	24.3	24.0	22.3	19.1	14.8	12.1
		MCDB	13.4	14.4	18.9	22.8	25.9	28.9	30.1	28.9	26.0	22.8	18.2	14.4
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.8	12.5	12.9	11.7	11.3	10.8	10.6	10.8	12.2	12.8	11.2
		MCDBR	14.0	14.5	14.8	15.5	13.6	13.0	12.6	12.7	13.6	14.2	14.9	13.3
		MCWBR	9.9	9.4	7.8	6.7	5.4	4.5	4.1	4.2	5.0	7.0	9.1	9.6
	5% WB	MCDBR	12.2	13.3	12.7	12.4	11.2	11.2	11.0	10.5	10.2	10.5	11.5	11.2
		MCWBR	9.8	10.0	7.8	6.0	4.8	4.5	4.1	4.1	4.4	6.1	8.6	9.2
Clear-Sky Solar Irradiance	taub	0.307	0.324	0.356	0.397	0.442	0.486	0.528	0.518	0.416	0.376	0.325	0.310	
	taud	2.509	2.456	2.406	2.292	2.199	2.117	2.020	2.030	2.311	2.401	2.505	2.534	
	Ebn at Noon	894	916	912	886	847	807	770	768	838	844	865	868	
	Edh at Noon	81	96	110	130	145	157	172	167	120	100	81	75	
All-Sky Solar Radiation	RadAvg	2.55	3.22	4.26	5.34	5.91	6.35	5.93	5.41	4.50	3.67	2.79	2.18	
	RadStd	0.17	0.39	0.30	0.41	0.48	0.47	0.39	0.27	0.41	0.47	0.22	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+100

Nomenclature: See separate page